

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Department of Computer Science
University of Pretoria

COS750 - Educational Software Development

Examination Specification

22 October 2026

1 Introduction

Data structures and algorithms are fundamental to computer science. Every degree programme that wants to be labelled as a computer science programme, therefore must include them in its curriculum. The core aspects of the *Algorithmic Foundations (AL)* knowledge area, provided in CS2023 [Kumar et al., 2024], are therefore the bare minimum for inclusion into a computer science degree programme.

You have been approached by the COS 212 – Data Structures and Algorithms – lecturer at the University of Pretoria to help develop an "app" to provide tutor-style, pre-lecture opportunities for students in the module.

2 Instruction

Design and develop a prototype of the "app" (to be referred to as the *COS212 app*) for one of the core data structures listed in the *Foundational Data Structures and Algorithms* knowledge unit (excluding one-dimensional arrays and one-dimensional linked lists), and one algorithm that can be applied to the data structure that is mentioned in CS2023 as core (from either CS Core or KA Core). The design and development of the prototype of the *COS212 app* will need to be completed before a presentation and demonstration of the app which will be evaluated during the scheduled exam time. The presentation should include the rationale for the design of the *COS212 app* and a demo of the app. A self-reflection report will further contribute to the examination for the module.

The *COS212 app*, and the presentation and the demonstration of the app will be completed in groups of 5 or 7. The self-reflection report will be completed individually.

2.1 *COS212 app* Prototype

Your group will need to choose one data structure and an accompanying algorithm from CS2023, and design and implement a prototype of the *COS212 app* for the chosen Data Structure and Algorithm. You cannot assume that students using your app have any knowledge of the data structure or algorithm as the app will be used prior to a lecture or discussion on the data structure and algorithm. The app should include some form of assessment that can be used to provide the student with additional scaffolding if they are not comfortable with the work, before attending the lecture or discussion.

Make use of a Software Engineering (SE) approach to design your *COS212 app*, and one of the Instructional Design (ID) approaches when designing the content for your app. Make sure you document both the SE and ID approaches in separate documents. These documents must be made available at the presentation and demonstration session.

The ID documentation should further reflect on your instructional choices. Which learning theories are to be applied, the learning styles the app is going to cater for, how the app is going to keep the student motivated, assess the progress of the student on the appropriate level *etc.* Effectively, applying everything you have learnt and will be learning in this module.

You may make use of Generative AI (GenAI) to help generate the content for your app and if you need, the GUI. Please acknowledge that you made use of GenAI and include your prompt and the tool you used in your ID documentation .

2.2 Presentation and Demonstration

During the examination period, your group will be need to be available for at most 2 hours, on either 16 or 17 November, to present your *COS212 app* to the examination panel and at least 2 peer groups who will be evaluating. Each group will be given a 30min slot. You should take no more than 10min for your presentation, 10min for your demonstration, 5 minutes for questions, and 5 minutes to setup and vacate the podium for the next team. All team members must be present for the presentation and demonstration.

3 Submission

Date issued: 22 October 2026

Updates issued: None

Deliverables: There are five deliverables:

1. A **Group registration** document with the name, student number, email and telephone number of each group member. Remember groups may only comprise 5 or 7 members. Provide a name for your group and the most favourite Smartie colour in your group.
2. **Prototype documentation**, which focusses on the development of the app for teaching the chosen data structure and the accompanying algorithm. Two documents need to be produced.
   - The SE document, needs to capture the software engineering approach and design of the app.
   - The educational objectives of the app and supporting choices made when designing the presentation, content, assessments, etc for the app (ID document).
3. **Presentation and demonstration** which provides an overview of your educational design decisions and a live demonstration of your prototype.
4. An individual **Self-reflection report** that highlights and provides insights into your journey through the design, development, presentation, demonstration and interactions with fellow team members during the process of creating the prototype.
5. A **GitHub link to the code**.

Due dates: The due dates for each of the deliverables are given below:

1. **Group registration**: Upload your group registration document to ClickUP before 20:00 on 23 October 2026.
2. **Prototype documentation**: Submit the SE documentation and ID documentation on ClickUP on the morning of 16 November 2026 before 8:00.
3. **Presentation and demonstration**: Make sure your presentation slides are uploaded to ClickUP before your presentation and provide a link to a repository where your prototype source code available.
4. **Self-reflection report**: Upload your individual report to ClickUP before 20:00 on Wednesday 18 November 2026.

4 Assessment

You will be assessed on your SE and ID documentation (50%), your presentation and demo (20%) of the *COS212 app*, and your self-reflection report (30%).

References

A. N. Kumar, R. K. Raj, S. G. Aly, M. D. Anderson, B. A. Becker, R. L. Blumenthal, E. Eaton, S. L. Epstein, M. Goldweber, P. Jalote, D. Lea, M. Oudshoorn, M. Pias, S. Reiser, C. Servin, R. Simha, T. Winters, and Q. Xiang. *Computer Science Curricula 2023*. Association for Computing Machinery, New York, NY, USA, 2024. ISBN 9798400710339.